

Cross-Layer Analysis of Thermal Tuning Stalls in Wafer-Scale Optical Interconnects for LLM MoE Training

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Mixture-of-experts (MoE) training is dominated by all-to-all communication, and wafer-scale optical interconnects based on dense wavelength-division multiplexing (DWDM) promise the bandwidth density it needs. Their microring resonators (MRRs), however, are held on resonance by thermo-optic tuning loops, and the compute bursts of MoE training move the temperature of the photonic layer by several kelvin within milliseconds. We quantify what this costs with a cross-layer analysis. The per-device compute timelines of a packet-level network simulation drive an Ansys transient thermal model of a 3D stack of a GPU, an electronic integrated circuit (EIC), and a photonic integrated circuit (PIC). The resulting resonance drift is converted into a per-round communication stall through the tracking loop and the detuning budget of the ring. The stalls are then fed back into the network simulation until the timeline and the stall agree. Die-temperature measurements on an H100 GPU match the modeled swing of 300 ms compute bursts within 15% and confirm the 700 W power level of the model. For Mixtral 8×7B and LLaMA-MoE 6.7B on a wafer fabric whose stall-free iteration time is within 3% and 16% of an ideal non-blocking fat-tree, a tracking loop limited to the 5 nm/s measured for an integrated implementation lengthens the iteration by 1.58× and 2.82× in an open-loop estimate. Feeding the stall back into the thermal model reduces this to 1.32× and 1.55×, because the stall itself lowers the duty cycle of each GPU and with it the temperature swing. The stall persists at full model depth, where it lengthens the iteration by about 1.16× and 1.25×. We then turn the penalty into specifications. The stall disappears for a loop that slews at 40 nm/s or reacts within 0.5 ms, for a heater pre-driven within about 1 ms of each GPU kernel launch, or for a dummy load that holds each GPU above 75% of peak power, whereas added thermal mass does not remove it. Removing the drift at the ring is the least costly option. An athermalized lithium-niobate ring with a non-volatile ferroelectric setpoint needs no holding power, no fast tracking loop, and no signal from the GPU, and it keeps its residual drift inside the budget provided its athermal point lies within about 3 K of the operating temperature. For a wafer fabric provisioned well enough to approach a non-blocking network, the tracking capability of the tuning loop is a first-order requirement.

CCS Concepts: • **Hardware** → **Emerging optical and photonic technologies**; **Interconnection networks**; • **Computing methodologies** → *Distributed computing methodologies*.

Additional Key Words and Phrases: mixture-of-experts, optical interconnects, wafer-scale systems, microring resonators, thermo-optic tuning, cross-layer co-simulation

1 INTRODUCTION

The rapid growth of large-scale artificial intelligence (AI) models has substantially increased the communication demands placed on modern computing systems. Large language models (LLMs) are trained on distributed accelerator platforms whose performance is increasingly limited by data movement as well as computation. The trend is amplified in mixture-of-experts (MoE) models, which add expert parallelism to data, tensor, and pipeline parallelism and therefore generate intensive all-to-all traffic, making the interconnect fabric a first-order design constraint.

Unlike dense transformers, MoE models route each token to a few selected experts, so their communication is bursty and non-uniform. Expert communication occupies a large fraction of the iteration time, and skewed expert

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selection concentrates traffic and computation on a subset of accelerators. MoE performance is therefore sensitive to link bandwidth, latency, and any overhead that delays a communication round.

The pressure is greatest in the scale-up domain, where accelerators within a node or package must exchange data at high throughput and low latency. Electrical interconnects face growing constraints in bandwidth density, energy efficiency, and reach as systems grow, and optical interconnects have emerged as the alternative, as reflected in studies of waferscale silicon photonics [4] and in optical interposer bridges for AI accelerators [2].

A key advantage of optical interconnects is dense wavelength-division multiplexing (DWDM), which carries many wavelength channels on one waveguide [39] and scales bandwidth without a proportional increase in the number of physical links. Packaging platforms such as TSMC’s COUPE place photonic interfaces directly next to processors [13], and system-on-wafer integration places whole wafers of compute dies on one substrate [34].

Bandwidth alone does not settle the question. DWDM links select, modulate, and route wavelengths with microring resonators (MRRs), which must be held on resonance, typically by a heater under closed-loop control [33]. MRRs are sensitive to temperature [7], and in a wafer-scale system the photonic layer sits directly beneath GPUs whose power swings by hundreds of watts between compute bursts and idle intervals. Whenever the ring drifts faster than its tuning loop can follow, the link stops carrying data until the loop catches up, and a fabric with ample nominal bandwidth can still deliver its data late.

How much this matters has not been quantified for real training workloads. Prior studies characterize thermal transients at the device level or use analytical workload models, and none couples the transients to the timing of the communication rounds, which decides whether a drift costs anything. We therefore close the loop between the layers. The per-device timelines of a packet-level network simulation drive a transient thermal model of the stack, the resonance drift sets a per-round communication stall, and the stalls are fed back into the network simulation until the timeline and the stall agree. We cross-check the thermal model against measurements on a real GPU, and we translate the result into what a tuning loop, a predictive heater drive, a dummy load, or a drift-free ring would each have to deliver.

The main contributions of this paper are as follows.

- **A closed-loop cross-layer framework, cross-checked on hardware.** A packet-level network simulation, a transient thermal model grounded in a published characterization of 2.5D and 3D co-packaged optics [8], and an optical-domain stall criterion are chained so that every delay is derived rather than assumed, and the chain is solved self-consistently. Die-temperature telemetry of an H100 GPU under synthetic and real MoE workloads matches the modeled swing of 300 ms bursts within 15%, confirms the 700 W power level of the model, and exposes a second, package-level drift timescale.
- **The tuning stall is a first-order, thermally coupled effect.** On a wafer fabric within 3% and 16% of an ideal fat-tree for Mixtral 8×7B and LLaMA-MoE 6.7B, a tracking loop at the measured 5 nm/s costs 1.58× and 2.82× in an open-loop estimate and 1.32× and 1.55× once the stall is fed back. Capturing it requires the thermal feedback, because the stall reshapes the duty cycle of each GPU and thereby its own thermal cause. Each collective waits for the device whose rounds fall in its cooling transients.
- **Quantified requirements for avoiding the stall.** A loop slewing at 40 nm/s or reacting within 0.5 ms, a heater pre-driven within about 1 ms of each GPU kernel launch, or a dummy load above 75% of peak power each removes the stall, and added thermal mass does not. The fabric itself needs a co-packaged-optics (CPO) port on every reticle and expert-parallel groups placed within a wafer.

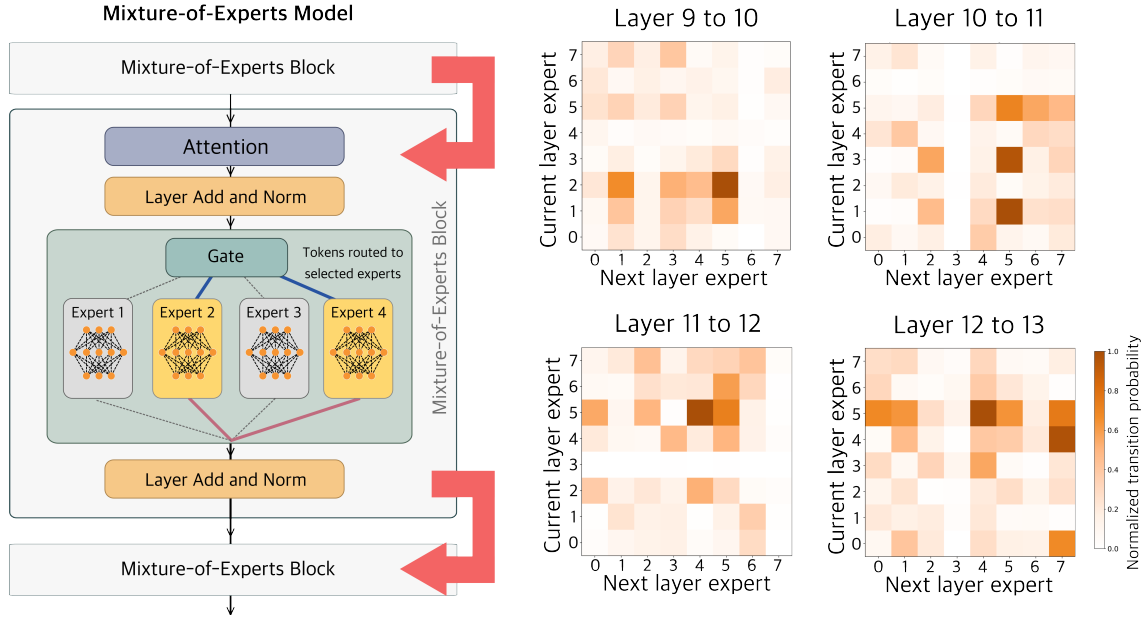


Fig. 1. Overview of an MoE block and expert-selection behavior in Mixtral 8x7B. The left diagram illustrates the computation flow within an MoE layer, including gating, expert routing, and aggregation. The right heatmaps show normalized transition probabilities from the expert chosen at one layer to the expert chosen at the next, for layers 9 to 13. A few transitions dominate, indicating non-uniform and persistent routing.

- **An athermal ring with a non-volatile ferroelectric setpoint as the least costly mitigation.** Compared with the other options, it is the only one that removes the stall with no holding power, no fast loop, no signal from the GPU, and no dependence on the workload. We quantify how close to the operating temperature its athermal point must lie and show why a ferroelectric actuator cannot instead track the drift.

2 BACKGROUND

2.1 Mixture-of-Experts Models and Sparse Activation

Mixture-of-experts (MoE) models scale large language models without increasing computation in proportion to the total parameter count. Dense transformers activate all feed-forward parameters for every token, whereas MoE models replace the feed-forward network (FFN) with a set of parallel experts and activate only a few of them for each token. Sparse activation lets the model capacity grow while the computation per token stays close to that of a much smaller dense model.

A typical MoE block consists of an attention layer, a gating function, and several parallel experts. After attention, the gate computes routing scores and selects the highest-ranked experts for each token. The selected experts process the token independently, and their outputs are combined into the block output. This data-dependent routing is the key architectural distinction between dense transformers and MoE models.

Expert selection is not uniformly distributed in practice [31]. Routing decisions follow structured patterns across tokens and layers, so some experts are used far more than others. Fig. 1 shows layer-to-layer expert transition probabilities for Mixtral 8×7B. A few transitions dominate, so some experts, and the devices that host them, receive a disproportionate share of the tokens.

2.2 Distributed MoE Training and Collective Communication Characteristics

Training modern LLMs requires distributed execution because parameters, activations, and optimizer states exceed the capacity of a single device. Models are parallelized by combining data parallelism (DP) [19], tensor parallelism (TP) [35], and pipeline parallelism (PP) [25], and MoE models add expert parallelism (EP) [18], in which different experts are placed on different accelerators.

Expert parallelism follows directly from the gated structure of the MoE block. Once the gate selects the active experts, token representations must be dispatched to the devices that host them, and after expert computation the outputs must be returned. Each MoE block therefore introduces two all-to-all phases, token dispatch before the experts and token combine after them, a pattern quite different from the structured collectives of dense-model training.

In our measurement of a four-layer Mixtral-style MoE block trained on eight H100 GPUs (Section 4.2), the dispatch and combine phases occupy 44% of the iteration on average and up to 57%. In the simulated wafer-scale configurations of this paper each device spends about half of the iteration waiting on all-to-all rounds or on the pipeline rather than computing, and this alternation drives its temperature swing.

The communication demand is also irregular. It varies over time, is non-uniform across experts, and concentrates on a subset of GPU pairs. MoE workloads are therefore sensitive to interconnect bandwidth, latency, and any per-round overhead, which motivates fabrics that support sparse, high-bandwidth all-to-all communication more efficiently than conventional electrical interconnects.

2.3 Communication Hierarchy and Scale-Up Bottleneck

Datacenter systems for AI are organized into a communication hierarchy of intra-chip, intra-package or scale-up, inter-node or scale-out, and inter-datacenter domains. The scale-up domain enables tightly coupled accelerator cooperation, and its demand grows rapidly with model size and parallelism degree, often faster than compute capability.

Existing scale-up interconnects are predominantly electrical, relying on high-speed serial links and proprietary fabrics. They face fundamental limits in bandwidth density because electrical I/O bandwidth scales with the perimeter, or beachfront, of a chip [2], whereas compute capability scales with its area. This growing imbalance motivates interconnect technologies with higher bandwidth density.

2.4 Optical Interconnects and Bandwidth Scaling via DWDM

Optical interconnects offer a path beyond the bandwidth-density limits of electrical links. In a DWDM system, many wavelength channels share one waveguide, so bandwidth scales with the number of wavelengths rather than with the number of physical interconnects. A single waveguide can carry tens of high-speed channels [39].

Advances in silicon photonics now allow modulators, waveguides, and photodetectors to be integrated closely with electronic circuits. Two approaches have emerged for AI systems, co-packaged optics (CPO), in which optical engines sit next to the processor package [6], and optical interposers, in which photonic components are integrated beneath compute and memory chiplets. Optical interposers provide a two-dimensional optical I/O surface, so bandwidth scales with chip area rather than chip perimeter.

2.5 Silicon Photonic Modulators and Design Trade-Offs

Silicon photonic interconnects encode electrical data onto optical carriers with electro-optic modulators. Mach-Zehnder interferometers (MZIs), microring resonators (MRRs), and electro-absorption modulators (EAMs) offer different trade-offs in footprint, thermal robustness, power, and optical bandwidth [42]. MZIs are broadband and thermally robust but long, EAMs are compact and broadband [3, 42], and MRRs are compact and wavelength selective, so each ring acts on one channel of a DWDM comb. This wavelength selectivity makes MRRs the natural device for DWDM links with 32 wavelengths per waveguide, and it is also what makes their thermal behavior central.

MRR operation, however, depends on precise resonance alignment, and practical implementations use thermal tuning to hold the resonance on its optical carrier. The resonance of a silicon MRR shifts strongly with temperature [28], so in tightly integrated accelerator systems the thermal fluctuations caused by computation shift their resonances and require continuous retuning. DWDM links may therefore be limited not only by their nominal bandwidth but also by how quickly their rings can be kept aligned.

3 WAFER-SCALE PHOTONIC SYSTEM ARCHITECTURE

As shown in Fig. 2, the architecture distributes compute and memory reticles across a wafer-scale photonic fabric that provides high-bandwidth optical communication throughout the system.

3.1 3D Integration of Compute and Memory Reticles

The system uses 3D-integrated compute and memory reticles (Fig. 3) that combine electronic and photonic components. Each reticle is formed by a hierarchical flow of wafer-to-wafer (W2W) and die-to-wafer (D2W) bonding [43]. The compute or memory die and its electrical integrated circuit (EIC) are first integrated with a photonic integrated circuit (PIC) through W2W hybrid bonding, and the resulting stack is then attached to the wafer-scale optical interposer through D2W bonding. This two-step scheme gives dense local electrical integration while connecting each reticle to the global optical fabric.

The top layer contains the GPU or memory die. Beneath it, the EIC implements the electrical front end for optical communication, including serializer and deserializer (SerDes) blocks, modulator drivers, transimpedance amplifiers (TIAs), equalizers, and link management circuits. The PIC implements the optical functions of the reticle, including waveguides, modulators, photodetectors, and wavelength-selective microring resonators [9]. The PIC couples to passive silicon nitride (SiN) waveguides in the interposer through vertical evanescent couplers based on inverse tapers [41].

3.2 Wafer-Scale SiN Photonic Interconnect

The reticles communicate through a wafer-scale SiN photonic interposer spanning a 300 mm wafer. The waveguide platform is based on reticle-stitched SiN routing [44], in which long waveguide sections are patterned over multiple reticles and stitched with high alignment precision, giving continuous optical paths of several tens of centimeters with low propagation loss and negligible stitching penalty.

To support many waveguide crossings, the architecture adopts a multilayer SiN network [14, 32] in which orthogonal routing directions are assigned to different SiN layers, with high-precision SiO₂ planarization enabling vertical transitions and crossings. The interconnect is thus organized as an orthogonal grid of row and column waveguides. Reticle-to-reticle communication uses optical-electrical-optical (OEO) relay rather than on-wafer optical switching, so

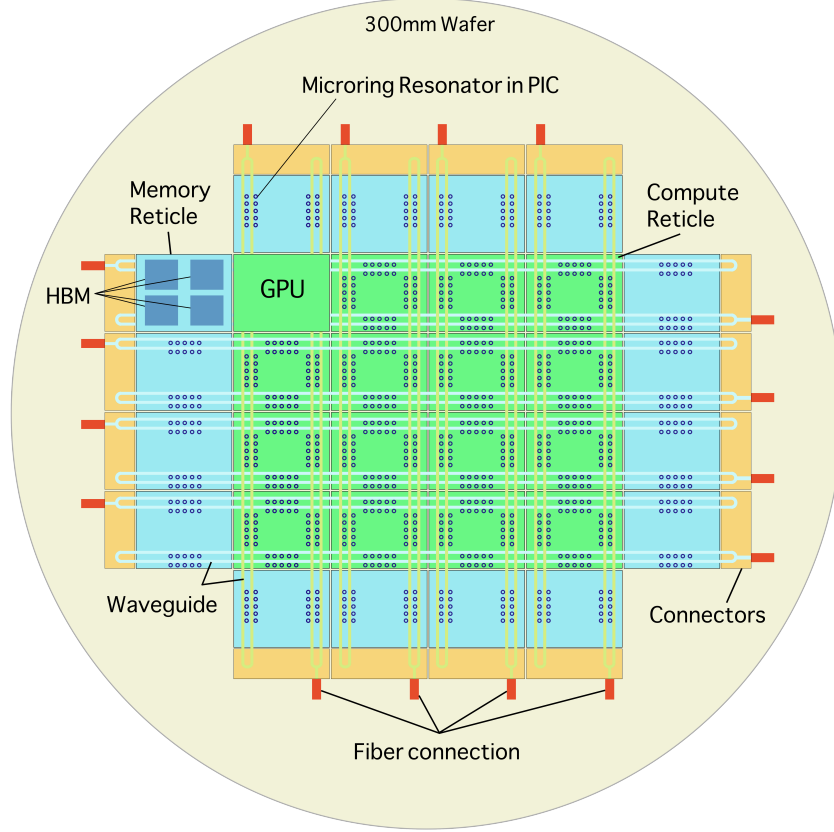


Fig. 2. Overview of the wafer-scale photonic system architecture. A 300 mm wafer integrates compute reticles (GPU chiplets) near the center and memory reticles (HBM) around the periphery. The photonic interconnect layer uses a grid of SiN waveguides for global communication, while microring resonators (MRRs) in the PIC layer provide wavelength-selective coupling and modulation. Off-wafer fibers are shown schematically; at the design point each reticle carries its own CPO port (Section 3.2).

that at an intermediate reticle the local EIC converts the incoming optical signal to the electrical domain and the local PIC retransmits it into the next waveguide segment. This avoids on-wafer optical switches and regenerates the signal.

Following prior photonic-interposer assumptions, we use 32 Gb/s per wavelength and 32 wavelengths per waveguide, i.e., 1.024 Tb/s per waveguide [14]. To reach 1.5 TB/s (12 Tb/s) of die-to-die bandwidth per direction, the required number of wavelength channels is

$$N_{\lambda} = \left\lceil \frac{12 \text{ Tb/s}}{32 \text{ Gb/s}} \right\rceil = 375,$$

and with 32 wavelengths per waveguide the required number of waveguides is

$$N_{\text{WG}} = \left\lceil \frac{375}{32} \right\rceil = 12,$$

which provides 12.288 Tb/s. Each wavelength needs one transmitter-side and one receiver-side resonator, so the end-point interface uses 375 of each, or 750 in total (768 when all 12×32 wavelength slots are provisioned).

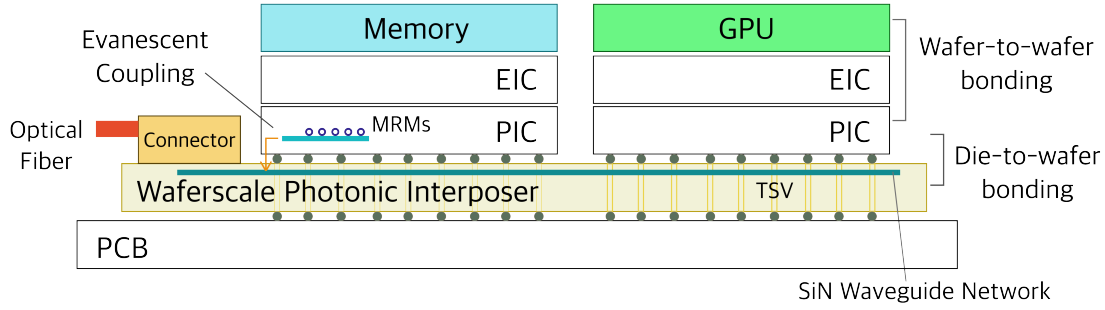


Fig. 3. Cross-sectional view of the 3D-integrated optical-interposer system. GPU and memory dies are vertically integrated with EIC and PIC layers through wafer-to-wafer bonding and then attached to a wafer-scale photonic interposer through die-to-wafer bonding. Optical signals couple into the SiN waveguide network through vertical evanescent couplers, while MRRs in the PIC provide wavelength-selective operation. Off-wafer fibers are shown schematically; at the design point each reticle carries its own CPO port (Section 3.2).

The architecture also provides optical interfaces for off-wafer connectivity. Each reticle stack carries a co-packaged-optics (CPO) port whose fibers leave the wafer through integrated connectors, so that traffic between wafers enters and leaves the fabric at the reticle that sources or sinks it rather than through a shared gateway. We model the off-wafer network that joins these ports as non-blocking, and Section 4.4 shows that per-reticle ports, rather than a shared gateway, let the fabric keep pace with a non-blocking electrical network.

3.3 Photonic Tuning Mechanisms

MRRs are aligned in conventional silicon photonics by thermo-optic tuning, in which a heater changes the refractive index of the ring. A control loop in the EIC monitors the ring and adjusts the heater, and its speed is bounded by the signal-to-noise ratio and update rate of its monitor rather than by the heater itself, whose time constant is microseconds. A step of the compute-die (XPU) power from 0 to 100% produces a thermal gradient of 1.78 K/ms, which for a 10 pm stabilization accuracy implies a controller reaction time of 84 μ s [7], whereas an integrated drop-port locking loop reports a tracking speed of 5 nm/s (0.0625 K/ms) [1]. A heater can only add heat, so the ring must be biased above its coldest operating point by the full expected swing.

An alternative actuator is a lithium niobate (LiNbO_3 , LN) ring with a hafnium zirconium oxide (HZO) ferroelectric gate stack [45]. An applied field changes the LN refractive index through the Pockels effect, and when the field exceeds the coercive field of HZO [23] the polarization switches and remains after the field is removed, so the resonance offset persists without holding power. With one ferroelectric gate the reported device shifts its resonance by about 60 pm in six non-volatile states, and with three gates placed around the ring by about 120 pm in 16 states, written with 10 μ s pulses at 65 fJ per state and cycled more than 10^7 times [45]. The HZO film loses less than 15% of its polarization below 90 $^\circ\text{C}$ [45], and the characteristic switching time of HZO is 5.4 ns [37]. We used technology computer-aided design (TCAD) to obtain the refractive-index distribution induced by the field in the HZO-gated LN waveguide and imported it into Ansys Lumerical to confirm that the TE and TM modes near 1550 nm remain guided under the applied field (Fig. 4). A bare LN ring drifts by roughly 25 to 50 pm/K, which we derive from the bulk thermo-optic coefficient of LN [24]. Cladding it with TiO_2 , whose thermo-optic coefficient is negative, cancels the first-order temperature dependence

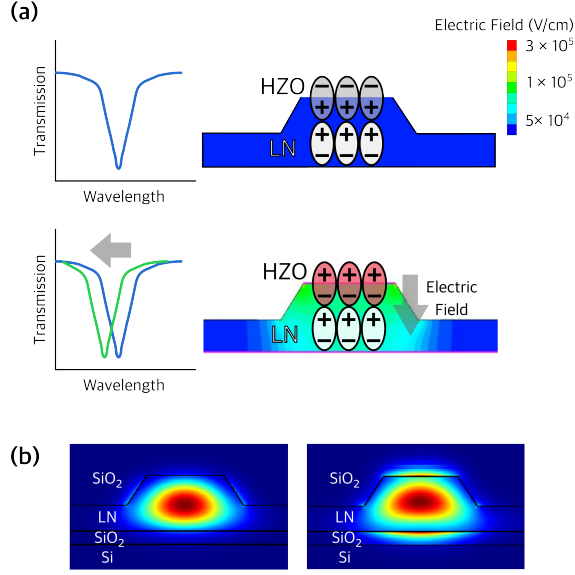


Fig. 4. Device-level modeling of a ferroelectric LiNbO₃ (LN) waveguide with an HZO gate stack. (a) Polarization switching under an applied electric field induces a non-volatile refractive-index change in the LN waveguide through the Pockels effect, shifting the transmission spectrum. (b) TE and TM mode profiles of the LN rib waveguide at the tuned wavelength, obtained in Ansys Lumerical using the TCAD-derived refractive-index distribution.

Table 1. LLM MoE training configurations used in the network and thermal simulations.

Configuration	Mixtral 8×7B	LLaMA-MoE 6.7B
MoE blocks in the full model	32	32
MoE blocks per pipeline stage in the simulation	1	1
Number of experts	8	16
EP degree	8	16
TP degree	4	1
PP degree	4	4
DP degree	2	2
GPUs (wafers of 16)	256 (16)	128 (8)
GPUs per expert-parallel group (wafers spanned)	32 (2)	16 (1)
Sequence length	4096	4096
Micro-batches per iteration and data-parallel replica	4	4

at a design temperature. The measured residual is within about ± 1 pm/K across 1480 to 1630 nm [11], and the second-order coefficient is 0.37 pm/K² with the athermal point at room temperature [21]. Section 4.8 examines what these two devices can and cannot do against the drift quantified below.

4 CROSS-LAYER THERMAL-TO-NETWORK EVALUATION

4.1 Evaluation Setup

Workload and network model. Mixtral 8×7B [16] and LLaMA-MoE 6.7B, the 16-expert LLaMA-MoE with 6.7B total parameters [46], are profiled on an NVIDIA H100 80 GB GPU with FlexFlow [15], using the task-graph export of the MixNet artifact [20], which records the forward and backward cost of every operator (attention, gating, expert feed-forward, all-to-all dispatch and combine, add-and-normalization) and exports the dataflow graph of one training iteration across all pipeline, expert-parallel, and data-parallel shards. The same graph drives the packet-level network simulation in htsim [12], which routes tokens according to the measured expert-selection matrix, and, through the per-device compute timelines that the simulation produces, the transient thermal analysis. Table 1 lists the parallelization settings. Each expert-parallel group spans two wafers for Mixtral and one for LLaMA-MoE, which matters because all-to-all traffic that leaves a wafer must cross the off-wafer ports. One correction is applied to the exported graphs. FlexFlow assigns zero bytes to the dispatch all-to-all (Group_by) operators, so we size each dispatch round like the combine round of the same layer, micro-batch, and direction, since both move the same token payload; the H100 measurement of Section 4.2 confirms equal byte counts in both directions.

Wafer fabric parameters. The simulated wafer implements Section 3.2 with a 4×4 array of GPU reticles per 300 mm wafer, an orthogonal row and column SiN waveguide grid, and OEO relay at row and column intersections. Each GPU owns 12 waveguides of 32 wavelengths at 32 Gb/s, i.e., 1.5 TB/s of egress. It has three row peers and three column peers, and each of these six links uses two waveguides, or 256 GB/s. Off-wafer connectivity uses one CPO port of 800 GB/s per reticle (Section 4.4). A non-blocking fat-tree and a direct all-to-all (“flat”) fabric serve as baselines at the same 1.5 TB/s per-GPU injection bandwidth.

GPU power model. The per-device timeline from the network simulation is translated into a time-varying XPU power profile by assigning each compute phase a power level according to its compute regime. We use a FLOP/byte roofline over the model dimensions (hidden size H , sequence length S , expert count E) with the H100’s FP32 throughput (67 TFLOPS) and HBM3 bandwidth (3.35 TB/s) [26], whose ratio gives a ridge point of about 20 FLOPs/byte; the classification below is the same with the bf16 tensor-core throughput. Attention and expert feed-forward general matrix multiplications (GEMMs) have arithmetic intensities of order S and receive the full 700 W XPU design power. The gate GEMM ($[S \times H] \rightarrow [S \times E]$) has an intensity of order $E/2$ and receives 40%. Elementwise add-and-normalization is memory-bound and receives 20%. Any interval in which the device executes no kernel, because it waits on an all-to-all round or on the pipeline, receives 12%. These fractions are our modeling choice. The 700 W level matches the 692 W that the H100 draws under sustained GEMM load, the board idles at 131 W (19%) in our measurement, and Section 4.9 shows how the stall depends on the idle level.

4.2 Thermal Model Grounding and Hardware Cross-Check

We derive the boundary conditions and material parameters of the transient thermal model from Coenen et al.’s CFD- and FE-based characterization of 2.5D and 3D co-packaged optics [8]. The modeled stack follows the OIO3D configuration of Coenen et al., in which the XPU (the compute die, here a GPU) sits on top, followed by a hybrid-bonded EIC and an active photonic integrated circuit (aPIC) that hosts the modeled MRRs, die-to-wafer bonded to a passive photonic interposer. Table 2 lists the parameters and their sources. The EIC and aPIC are constant background heat sources at their reported power densities, the XPU receives the phase-resolved power trace of Section 4.1, and cooling is applied at the top surface of the XPU through a Si microchannel cold plate. The transient is solved with a fixed 1 ms

Table 2. Physical parameters used in the OIO3D transient thermal model, derived from Coenen et al. [8].

Parameter	Value	Source
Si microchannel cold-plate heat-transfer coefficient (top-side XPU cooling)	$7 \times 10^4 \text{ W m}^{-2} \text{ K}^{-1}$	Fig. 11 (conservative reference)
Effective hybrid-bond conductivity with vias (XPU to EIC and EIC to aPIC)	$5.0 \text{ W m}^{-1} \text{ K}^{-1}$	Table I ($R = 0.48 \text{ mm}^2 \text{ K W}^{-1}$, $t = 2.4 \text{ }\mu\text{m}$)
Effective D2W-bond conductivity without dense vias (aPIC to interposer)	$2.55 \text{ W m}^{-1} \text{ K}^{-1}$	Table I ($R = 0.94 \text{ mm}^2 \text{ K W}^{-1}$), applied by us
Interposer-to-substrate boundary resistance	0.1 K W^{-1}	Sec. II
EIC power density	0.426 W mm^{-2}	Sec. II-B
aPIC power density	0.113 W mm^{-2}	Sec. II-B
XPU power (GEMM phases)	700 W	Secs. II-B and IV-A
OIO3D tile footprint	$25 \times 25 \text{ mm}^2$	Sec. II

time step over ten repetitions of the simulated iteration so that the trace reaches its periodic steady state, and the first 1.6 s, which contain the start-up transient, are discarded. Automatic time stepping is not used, because it coarsened to 15 to 175 ms during long idle intervals and quantized every derived stall to that step.

Which device to model. Devices in one expert-parallel group do not share a timeline, because the order and timing of their compute and communication differ. We build the power schedule for the device with the median compute share and for the device with the smallest one, and inject the larger of the two stall estimates for every round type, because an all-to-all round completes only when its slowest participant can transmit. With per-reticle ports the per-device load is nearly uniform, and the duty cycles of the two devices agree within 0.1%, yet their stalls differ; for LLaMA-MoE the delay per round is 20 ms for the median device and 39 ms for the least-loaded one, because what matters is when each round falls relative to the cooling transients. Injecting the larger value for every device ensures that every round waits for its slowest participant.

Hardware cross-check. No workload-driven temperature measurement of an OIO3D stack exists, so we compare the model’s dynamics against a conventionally packaged GPU without fitting any model parameter to the measurement. On one H100 SXM node we ran synthetic burst and idle GEMM loops and a real four-layer Mixtral-style MoE forward and backward loop (eight GPUs, expert parallelism over NVLink, bf16) while sampling the die-temperature sensor, which reports in 1 K steps, every 20 ms (Fig. 5). Under sustained GEMM load the H100 draws 692 W, which confirms the 700 W level of the power model, and it idles at 131 W. Two timescales appear. The die responds to 307 ms bursts with a 13 K per-cycle swing and to 42 ms bursts with about 4 K, so the fast swing is limited by how far heat diffuses during the burst. Under 60 s bursts the die swings by 33.5 K with a time constant of 6 to 7 s on heating and 9 s on cooling. NVML reports power averaged over a window longer than the short bursts, so we take the burst power as the sustained 692 W. To compare like with like, we drive the thermal model of the stack, through a reduced-order model of two first-order lags fitted to its Ansys response (Section 4.9), with the same burst patterns between 131 and 692 W (Fig. 5(b)). For 307 ms bursts the model swings by 11.4 K against the measured 13 K, an agreement within 15%. For 42 ms bursts the model swings by 8.9 K, more than the die sensor reports, as expected for a stack without a lid that is read at the photonic layer rather than through a die sensor with 1 K resolution. For 60 s bursts the model reaches 11.4 K and the measurement 33.5 K, and the difference is the slow component of the lidded package, which the directly cooled stack does not have. The compute blocks of the simulated models arrive back to back within each busy block, so the stack accumulates swings of 11.4 and 10.2 K (Fig. 6). In the same stack under representative MoE schedules, adding

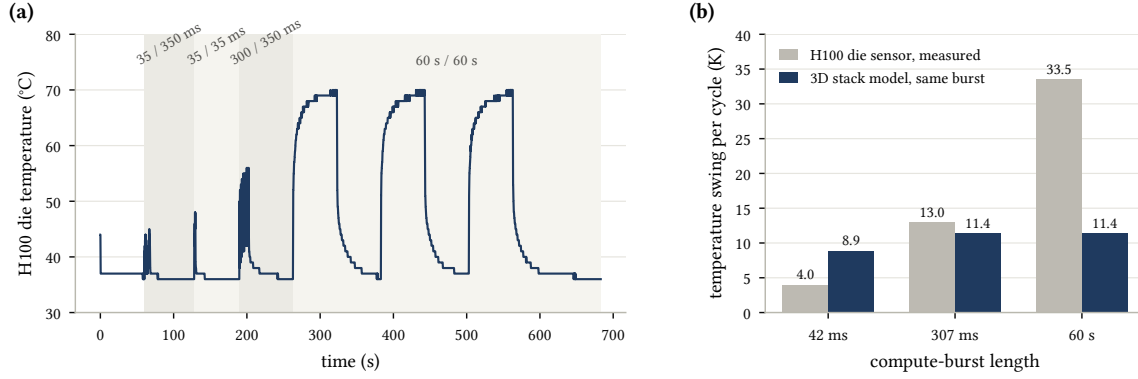


Fig. 5. Hardware cross-check on an NVIDIA H100 SXM GPU (die-temperature sensor of the NVIDIA management library, NVML, sampled every 20 ms). (a) Die temperature under four synthetic compute patterns with nominal bursts of 35 ms every 385 ms, 35 ms every 70 ms, 300 ms every 650 ms, and 60 s on and off. (b) Per-cycle temperature swing for the measured burst lengths of 42 ms, 307 ms, and 60 s, measured by the die sensor and computed for the 3D stack model under the same burst pattern between the measured idle of 131 W and the sustained GEMM power of 692 W. The two agree at 307 ms. For short bursts the lidless, directly cooled stack responds faster and more strongly than the lidded package, and for long bursts the package adds a slow component that the stack does not have.

3 or 10 mm of copper as cold-plate mass damps the swing by 17 to 45% and raises the operating floor by up to 7.7 K, because the millisecond-scale swing is set by diffusion inside the die. The measured MoE loop also shows that dispatch and combine move equal byte counts, with all-to-all phases occupying 44% of a 1.59 s iteration on average (24 to 57% across iterations).

4.3 From Temperature to Communication Stall

A temperature trajectory alone does not establish a communication outage. A fast cooling rate causes a stall only if the resulting resonance shift outruns what the tracking loop and the optical margin of the channel can absorb, so we convert $T(t)$ into per-round delays through an explicit chain.

First, the thermal shift of the ring resonance is $\Delta\lambda_{\text{th}}(t) = (d\lambda/dT) [T(t) - T(0)]$ with the silicon thermo-optic coefficient of 80 pm/K. Second, the control loop tracks this shift subject to its maximum wavelength slew rate R_{ctrl} . We adopt the measured tracking speed of the integrated drop-port locking loop of Agarwal et al. [1], which reports a tracking speed of 5 nm/s, a tuning range of 5 nm, and an accuracy better than 10 pm. At 80 pm/K the tracking speed is 0.0625 K/ms (5 pm/ms). We report every stall for this loop, and Section 4.7 generalizes the result by treating both the slew rate and the loop bandwidth as design variables. Third, the residual detuning $\varepsilon(t) = |\Delta\lambda_{\text{th}}(t) - \lambda_{\text{track}}(t)|$ is the wavelength error the link actually sees. Fourth, the link is stalled while $\varepsilon(t)$ exceeds an acceptable-detuning budget ε_{max} of 10% of the resonance full width at half maximum (FWHM). For a drop filter this budget is strict, since a Lorentzian loses 0.17 dB at that detuning and 1 dB at 25% of the FWHM, while for a ring modulator, whose operating point sits on the slope of the resonance, a tenth of a linewidth is the scale at which the modulation amplitude degrades. Section 4.9 shows that the stall depends only weakly on this choice up to 30% of the FWHM. We take a loaded quality factor Q of 8000, whose photon-lifetime bandwidth of 24 GHz supports 32 Gb/s signaling, so that the FWHM at 1550 nm is 193.8 pm and the budget ε_{max} is 19.4 pm.

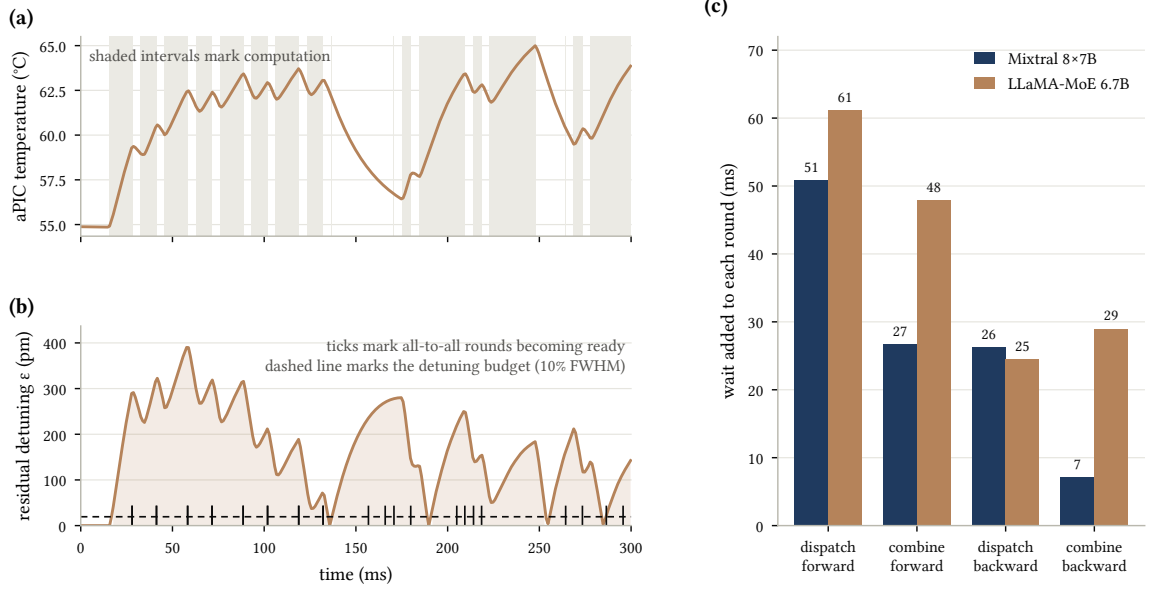


Fig. 6. From temperature to communication stall at the 800 GB/s-port design point. (a) aPIC (MRR) temperature during 300 ms of a steady-state LLaMA-MoE 6.7B iteration for the least-loaded device. Shaded intervals are compute bursts, and unshaded intervals are waits on all-to-all rounds or on the pipeline. (b) Residual detuning $\epsilon(t)$ after the rate-limited thermo-optic tracker. The dashed line is the 19.4 pm budget, and each tick marks an all-to-all round of this device’s layer becoming ready. Dispatch rounds become ready immediately after a burst, inside the cooling transient. (c) Wait added to each all-to-all round, by round type, for the two models. A round that becomes ready while its rings are out of the budget waits until they return, so dispatch rounds, which start at the beginning of a cooling transient, wait longest. These per-type values are the delays injected into the network simulation.

Finally, the stall is mapped onto communication. An all-to-all round that becomes ready while the ring is out of budget waits for the remainder of that stall window, and any stall that begins inside the round adds to its duration. This per-round delay, not a mean stall-window length, is what the network simulation injects before each round (Section 4.5), and Section 4.6 feeds the stalled timeline back into the thermal model.

Fig. 6 shows the chain at the design point. The aPIC swings by 11.4 K for Mixtral and 10.2 K for LLaMA-MoE between bursts and idle intervals, i.e., by 0.91 and 0.82 nm of resonance shift or more than four linewidths. The fastest cooling slopes reach 0.40 and 0.33 K/ms (median 0.09 and 0.12 K/ms), five to six times R_{ctrl} , and smaller than the 1.78 K/ms of a 0 to 100% XPU power step [7]. The residual therefore exceeds the budget for most of each busy block and returns inside it only briefly between bursts. What determines the communication penalty is where the all-to-all rounds fall relative to these windows. Dispatch rounds are issued immediately after the gating GEMM, at the start of the cooling transient, and combine rounds follow the expert GEMMs in the same way. The resulting delay per round, for the device class that stalls most, averages 28 ms for Mixtral and 39 ms for LLaMA-MoE. The per-type values we inject (dispatch and combine, forward and backward), taken as the larger of the two device classes, are 51, 27, 26, and 7 ms for Mixtral and 61, 48, 25, and 29 ms for LLaMA-MoE.

Two features of this result differ from a naive reading of the temperature traces. First, the device that stalls most is the one whose rounds fall at the start of its cooling transients. With nearly equal duty cycles, the phase of the rounds relative to the bursts, not the expert load, decides the stall. Second, the delay a round suffers is set by the stall window

Table 3. Stall-free iteration time in milliseconds for the simulated configurations of Table 1, with dispatch rounds sized like combine rounds. Every fabric gives each GPU 1.5 TB/s of injection bandwidth, and the wafer fabric with CPO ports additionally has its port rate for off-wafer traffic.

Fabric	Mixtral 8×7B	LLaMA-MoE 6.7B
Non-blocking fat-tree (ideal)	841.3	427.4
Direct all-to-all (“flat”)	1152.7	1560.1
Wafer, one 200 GB/s gateway link per wafer pair	1954.9	493.1
Wafer, 800 GB/s CPO port per reticle (design point)	865.3	494.4

it lands in rather than by its own length. At 1.5 TB/s per GPU the rounds of Mixtral and LLaMA-MoE last 0.4 and 4.5 ms, far shorter than the 50 to 60 ms stall windows, so a round that becomes ready inside a window waits for nearly all of it.

4.4 Network Baselines and the Design Point

Before the stall is quantified, the stall-free fabric is compared with conventional alternatives at equal per-GPU injection bandwidth, since comparisons at equal per-link rate favor whichever topology has more links per node. Table 3 gives the result. How the wafers are connected to each other dominates. With a single 200 GB/s optical gateway link between each pair of wafers, so that all traffic between two wafers is funneled through one reticle pair after extra intra-wafer hops, the wafer fabric is 2.3× slower than the ideal fat-tree for Mixtral, whose expert-parallel groups span two wafers. Giving every reticle its own CPO port, as Section 3.2 proposes, removes the funnel, and at 800 GB/s per port Mixtral’s iteration time falls from 1955 to 865 ms, within 2.9% of the ideal fat-tree. The 16-GPU expert-parallel group of LLaMA-MoE stays within one wafer, and LLaMA-MoE is 16% slower than the fat-tree because of the two-hop relay paths of the row and column mesh.

The ports add 0.8 TB/s of off-wafer capacity per GPU on top of the 1.5 TB/s on-wafer injection, but this does not bias the comparison, because a fat-tree given the same 2.3 TB/s per GPU runs Mixtral and LLaMA-MoE in 839.3 and 426.3 ms, within 0.3% of its 1.5 TB/s times. The flat fabric is 1.33× and 3.2× slower than the wafer fabric, because splitting 1.5 TB/s over up to 255 links leaves each link too narrow for the bursty rounds. Two design rules follow, namely a CPO port on every reticle rather than a shared gateway, and a placement that keeps each expert-parallel group within one wafer where the parallelization allows it. The wafer fabric thus matches an ideal non-blocking fat-tree closely while needing far fewer cables and transceivers, which the next paragraph quantifies.

Why wafer-scale. In per-GPU terms, each reticle injects 1.5 TB/s into the wafer through twelve waveguides, 3.3× the per-direction NVLink 4 bandwidth of an H100 SXM and 1.7× the per-direction NVLink 5 bandwidth of a Blackwell GPU, whose NVLink bandwidths are 900 GB/s and 1.8 TB/s bidirectional [27], and a further 0.8 TB/s off the wafer through its CPO port. On-wafer traffic needs no switch; it crosses one waveguide, or two with an OEO relay at a row and column intersection, and reaches every GPU of the wafer. Realizing 1.5 TB/s per GPU with a non-blocking fat-tree would take fifteen 800 Gb/s ports per GPU and, for the 256 GPUs of the Mixtral configuration built from 64-port 800 Gb/s switches, three tiers with 45 cables and 90 transceiver ends per GPU, whereas the wafer needs fourteen fibers per reticle for its port (seven per direction at 1.024 Tb/s each) and none for on-wafer traffic. This density is carried by the resonators, whose thermal behavior is the subject of the rest of this section.

Table 4. Iteration time with a 5 nm/s tuning loop, without and with the feedback of the stall onto the thermal field.

Case	Mixtral 8×7B	LLaMA-MoE 6.7B
No tuning stall	865 ms (1.00×)	494 ms (1.00×)
Without feedback (open loop)	1371 ms (1.58×)	1396 ms (2.82×)
With feedback (closed loop)	1140 ms (1.32×)	768 ms (1.55×)

4.5 Tuning Stall without Thermal Feedback

The per-round delays of Section 4.3 are injected into the network simulation as a one-time stall before each all-to-all round of the corresponding type, not as link propagation latency, which would compound on every packet, because a link recovers once its ring returns within budget, so the delay applies once per round. In this open-loop estimate the delays are derived from the stall-free timeline and are not fed back. With the 5 nm/s loop the stall increases the iteration time to 1.58× for Mixtral 8×7B (865 to 1371 ms) and 2.82× for LLaMA-MoE 6.7B (494 to 1396 ms). The ordering follows neither the temperature swing nor the communication volume. It follows the injected delay relative to the stall-free time, because the iteration serializes the rounds of its four micro-batches, so the added time scales with the summed per-type delays (111 and 163 ms per micro-batch), at about 4.5 and 5.5 times these sums, and LLaMA-MoE combines the larger delays with the shorter iteration.

4.6 Tuning Stall with Thermal Feedback

The open-loop delays come from a timeline that has no stall in it, whereas the stall changes the timeline. A stalled GPU waits longer, dissipates less heat, and swings less in temperature, which in turn shrinks the stall. We therefore feed the stalled timeline back. Its per-device compute schedules are rebuilt from the stalled network simulation, the thermal transients are recomputed, and new per-type delays are derived for the same devices. The physically consistent result is the self-consistent point, at which the delay injected into the network simulation equals the delay derived from the resulting timeline. Table 4 compares it with the open-loop estimate.

The feedback is strong. Injecting the open-loop stall stretches the iteration by 1.6 and 2.8 times, so the duty cycle of the modeled devices falls from 51% to 32% (Mixtral) and from 41% to 15% (LLaMA-MoE), and the delays derived from this stretched timeline fall to 8.3 and 5.5 ms per round on average, 70 to 86% below the open-loop values. Injected in turn, these overshoot in the other direction and give only 1.16× and 1.22×, and a further undamped step returns close to the open-loop delays, so the plain iteration oscillates. We therefore damp it, injecting at each step the mean of the previously injected and the newly derived delays. The damped iteration contracts, with the gap between injected and derived mean delay falling from 7.8 to 2.7 to 1.6 ms for Mixtral and from 15.4 to 3.7 to 0.8 ms for LLaMA-MoE, and after three damped steps the derived mean delay agrees with the injected one within 11% and 6%. At this self-consistent point the mean delay per round is 15 and 14 ms, the duty cycle of the modeled devices is 39% and 27%, and the tuning stall lengthens the iteration by 1.32× and 1.55×. The closed-loop excess is 54% and 30% of the open-loop excess, which shows that the coupling between the network timeline and the thermal field is itself a first-order effect that the self-consistent analysis resolves. The stall cannot remove itself, because reducing it restores the duty cycle and with it the swing that causes it.

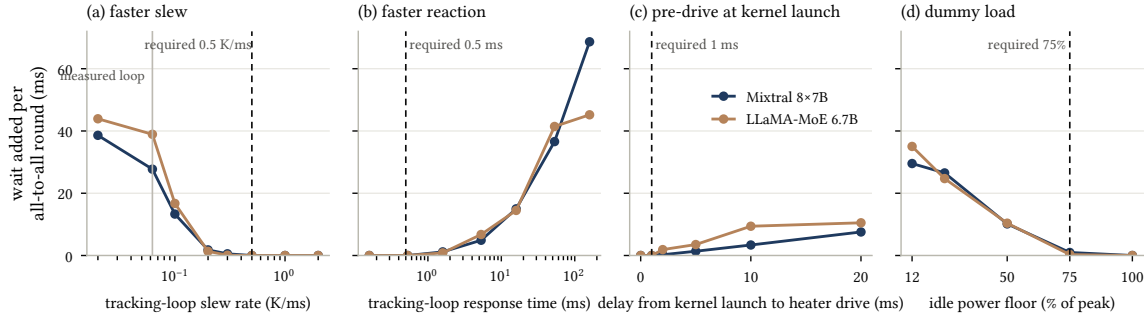


Fig. 7. Four ways to remove the stall and what each requires, on the open-loop traces. Each panel shows the mean wait added per all-to-all round as one design parameter varies, and the dashed line marks the value at which the wait falls to 1 ms or less. (a) Slew rate of the tracking loop; the solid grey line marks the measured 5 nm/s loop [1]. (b) Response time of a tracking loop without a slew limit; 0.5 ms corresponds to a loop bandwidth of 300 Hz. (c) Delay between a GPU kernel launch and the start of a predictive heater drive, at the measured slew rate. (d) Idle power floor of a dummy load held during waits, computed with the reduced-order thermal kernel.

4.7 Mitigation Requirements

Fig. 7 is a design chart with one panel per mitigation, from which a designer reads the requirement for a given implementation. All requirements in this subsection are evaluated on the open-loop traces, which have a higher duty cycle and swing than the closed loop of Section 4.6 and therefore bound it from above. The per-round delay falls to about 2 ms at 0.2 K/ms and to zero at 0.5 K/ms (40 nm/s), eight times the adopted value, for both models (Fig. 7(a)). A slew limit is one of two ways a loop can fall behind. The other is a slow reaction, in which the loop can move the heater as fast as needed but notices the drift late (Fig. 7(b)). We model this with a first-order tracking loop without a slew limit and characterize it by its response time τ , the inverse of 2π times its bandwidth. During the steepest cooling the resonance moves by about 32 pm per millisecond, and such a loop trails it by roughly the drift accumulated over one response time, so the error stays inside the 19.4 pm budget only if τ is below about 0.6 ms. The simulation confirms this. The wait per round is about 15 ms for a 16 ms response time, 5 to 7 ms for 5 ms, about 1 ms for 1.6 ms, and zero for about 0.5 ms, which corresponds to a loop bandwidth of 300 Hz. Reported loops track at 5 nm/s with a 5 nm tuning range [1], lock three rings over more than 40 °C from a single monitor [10], and tune over more than 50 K at 3.8 μ W/GHz with bit-statistics monitoring [38]. A dithering loop has stabilized a ring against 10 Hz thermal oscillations of 5 K [29], a drift rate of roughly 0.16 to 0.31 K/ms that approaches the 0.5 K/ms requirement. The requirement is therefore a specification for the EIC, and a modest one in loop bandwidth.

A second option is to pre-drive the heater at kernel launch. The GPU knows when each compute burst starts and ends, and the temperature follows from those events. If the GPU signals each kernel launch to the EIC, the controller can drive the heater along the temperature trajectory predicted by a thermal model and leave only the prediction error to the slow feedback loop. Fig. 7(c) shows how the result depends on the delay between the kernel launch and the start of the heater drive. Within 1 ms the wait per round stays at zero, at 2 ms it is up to 2 ms, and at 10 ms it is 3 to 9 ms. Replaying the trajectory of the previous iteration instead cannot meet this, since all-to-all durations measured on the eight-GPU H100 system vary by tens of milliseconds between iterations.

Table 5 collects these options with two thermal ones and the drift-free ring of Section 4.8. Holding the GPU at a dummy load during its waits removes the swing at its source (Fig. 7(d)). Raising the idle floor from 12% to 50% of peak

Table 5. Requirement and cost of each mitigation for a wait of 1 ms or less per all-to-all round for both models at the design point. Holding power is the continuous power the mitigation adds per GPU; the heater value is a lower bound at the most efficient reported heater (Section 4.8).

Option	Requirement	Holding power per GPU	Added control	Tied to workload dynamics
Faster slew	≥ 0.5 K/ms (40 nm/s), $8\times$ the measured loop [1]	≥ 0.5 W heater bias	fast tracking loop	yes, steepest cooling slope
Faster reaction	response time about 0.5 ms (300 Hz)	≥ 0.5 W heater bias	fast tracking loop	yes, steepest cooling slope
Heater pre-driven at GPU kernel launch	started within 1 ms of the compute burst	≥ 0.5 W heater bias	kernel-launch signal to the EIC, thermal predictor	yes, burst timing
Dummy load during waits	idle floor $\geq 75\%$ of peak	215 to 265 W	GPU power control	yes, idle intervals
Cold-plate thermal mass	not sufficient (swing reduced by 17 to 45%)	none	none	not applicable
Athermal LN ring, non-volatile setpoint	athermal point within 3 K of the operating temperature	none (65 fJ per write)	none	no

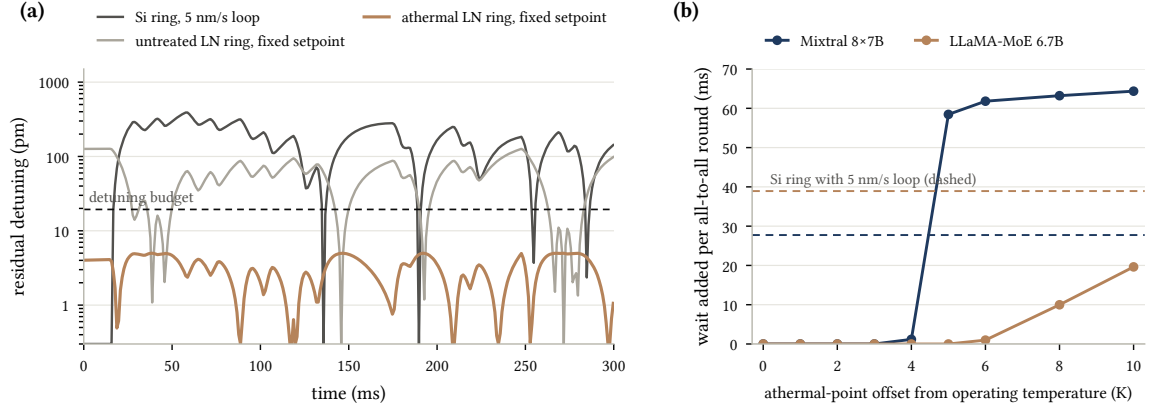


Fig. 8. Silicon ring versus lithium-niobate ring under the simulated workload. (a) Residual detuning during the LLaMA-MoE window of Fig. 6. The silicon ring drifts strongly and its loop follows too slowly, the untreated LN ring drifts less but is not followed at all, and the athermal LN ring barely drifts, so it needs no tracking loop. (b) Wait added per all-to-all round for an athermal LN ring with a fixed setpoint, versus the offset of its athermal point from the mean operating temperature.

power cuts the delay per round to about 10 ms, and 75% brings it to 1 ms or less, at 130 to 160 W and 215 to 265 W of additional average power per GPU. Thermal mass damps the swing by at most 45% and raises the operating point. The remaining option, a ring that does not drift, is examined in Section 4.8. The analysis thus gives each option a concrete specification.

4.8 Removing the Drift at the Ring

Instead of tracking the drift faster, one can reduce it at the ring. Section 3.3 introduced the two devices this requires, and here we test them against the thermal statistics of Section 4.3.

Drift versus budget. The steady-state aPIC swing is 11.4 K for Mixtral and 10.2 K for LLaMA-MoE. A silicon ring (80 pm/K) moves by 0.82 to 0.91 nm, four to five linewidths, and an untreated LN ring (25 to 50 pm/K, Section 3.3) by 255 to 570 pm, 13 to 29 times the 19.4 pm budget. For an athermalized LN ring the first-order term vanishes at the athermal point T_a and the residual is $a(T - T_a)^2$ with $a = 0.37 \text{ pm/K}^2$ [21]. Fig. 8(a) applies the three cases to the same temperature trace. The silicon ring with the 5 nm/s loop exceeds the budget after every burst, the untreated LN ring with a fixed setpoint exceeds it for most of the window, and the athermal LN ring with a fixed setpoint, its athermal point at the mean operating temperature, stays within about 7 pm, inside the budget with no tracking loop at all. Fig. 8(b) turns this into the wait per all-to-all round. The athermal ring adds no wait while its athermal point lies within 3 K of the mean operating temperature for Mixtral (61 °C) and within 5 K for LLaMA-MoE (60 °C), compared with 28 and 39 ms per round for the silicon ring. Beyond that offset the fixed setpoint no longer covers the swing, and for Mixtral the wait exceeds that of the tracked silicon ring, so the offset tolerance is a design limit and the cladding has to be designed for the operating temperature of the stack. A slower operating-point drift, such as that of a surrounding package, moves the athermal residual by tens of picometers at most and is absorbed by rewriting the non-volatile setpoint over seconds rather than milliseconds. The budget used here is that of a $Q = 8000$ ring, and a ring with a narrower linewidth has a proportionally smaller budget and tolerance.

Why a non-volatile element cannot track. The ferroelectric setpoint cannot follow the drift of an untreated LN ring, because its range of 60 to 120 pm is smaller than the 255 to 570 pm of that drift. Even within its range, following the drift would take one write each time the resonance moved by one budget width, about $2\Delta\lambda/\varepsilon_{\max}$ writes per compute burst over its heating and cooling legs, i.e., 26 to 59 writes per burst, or about 400 to 940 writes per iteration with 16 bursts per device per simulated iteration (four micro-batches, each with a forward and a backward pass of two compute blocks). The device has been cycled 10^7 times [45], as have HZO capacitors at 125 °C [23], and 10^7 writes at this rate last about 10^4 iterations. As a setpoint on an athermal ring, by contrast, the element is written at calibration and when the operating point changes, and even one write per 10 s over a 10^6 -iteration run at full depth is of order 10^6 writes, within the demonstrated cycling.

Power, range, and resolution. A thermo-optic heater can only raise the ring temperature, so it must be biased above the coldest operating point by at least the full 0.9 nm swing, and every one of the 750 on-wafer endpoint rings of Section 3.2 and the 400 rings of the CPO port (200 wavelengths per direction) holds that bias continuously. At the $3.8 \mu\text{W}/\text{GHz}$ of the most efficient reported heater [38], covering the 0.91 nm (114 GHz) swing costs at least 0.43 mW per ring, or 0.5 W per GPU for its 1150 rings, and less efficient heaters cost proportionally more. A ring whose bias runs out must relock to the next free-spectral-range order, a mode hop that is itself a millisecond-scale stall. The non-volatile setpoint dissipates no holding power, and each write costs 65 fJ [45]. Its resolution fits the budget. The six states of one gate are about 12 pm apart and the 16 states of three gates about 8 pm apart, so a ring programmed to the nearest state sits within 4 to 6 pm of its target, inside the 19.4 pm budget. A 15% polarization loss corresponds to 9 to 18 pm, also inside the budget, and a rewrite restores it. Its range of 60 to 120 pm is the constraint. It covers the athermal residual of about 7 pm and a slow operating-point drift of tens of picometers, while an as-fabricated resonance offset larger than the range calls for a one-time trim outside it. Where a finer or faster correction is wanted, the Pockels effect of the same LN waveguide offers it at the 7 pm/V reported for thin-film LN racetrack resonators [40], so half a state corresponds to about 0.6 to 0.9 V across a capacitive load, within the supply range of an EIC.

Comparison of the mitigations. Table 5 places the athermal ring with a non-volatile setpoint beside the other options. Every option can bring the wait per round to zero, so they differ in what they cost. The faster loop, the faster reaction, and the predictive heater drive all keep the silicon ring and its heater, and with it a holding power of at least

0.5 W per GPU, and they add either a tracking loop eight times faster than the measured one or a signal path from the GPU to the EIC with a thermal predictor. The dummy load needs no change to the ring but spends 215 to 265 W per GPU. Each of these four is also tied to the workload, since the loop must be designed for the steepest cooling slope, the prediction for the burst timing, and the dummy load for the idle intervals, so a different model or parallelization changes the requirement. The athermal ring with a ferroelectric setpoint is the only option that needs no holding power, no fast loop, and no signal from the GPU, and whose operation does not depend on the workload, because it removes the drift at its source rather than following it. Its requirement is on the device instead, namely an athermal point within about 3 K of the operating temperature and the integration of the TiO_2 cladding with the ferroelectric gate on one ring, which Fig. 8 and the analysis above quantify.

4.9 Sensitivity and Full Model Depth

Full model depth. In the full models each pipeline stage holds eight MoE blocks, so a device computes several blocks back to back before it waits on the other stages. Because the thermal model is linear, a reduced-order model of two first-order lags of 12 and 30 ms, with gains of 1.5 and 11.0 K at full power, fitted to one Ansys trace, reproduces the other traces within 0.3 K rms. Applied to full-depth schedules, it gives a wait of 15 ms per round for Mixtral and 11 to 16 ms for LLaMA-MoE, and a full Ansys transient of the Mixtral schedule over 41.5 s at 1 ms confirms it with a swing of 11.7 K and a wait of 14.7 ms per round. The long stalls that follow a wait on the other stages then occur once per eight blocks instead of once per block, while the stalls set by the swing inside a busy block remain, and with thermal feedback the stall still lengthens the iteration by about $1.16\times$ for Mixtral and $1.25\times$ for LLaMA-MoE.

Sensitivity to the ring and power assumptions. The stall depends on the quality factor Q of the ring, which sets the width of its resonance, and on the detuning budget $\epsilon_{\max}$, the drift a link tolerates before it stalls. Re-evaluating the open-loop per-round delays over $Q \in [5000, 15000]$ and $\epsilon_{\max} \in [5\%, 20\%]$ of the FWHM moves the means from 28 to between 25 and 37 ms for Mixtral and from 39 to between 27 and 53 ms for LLaMA-MoE. Widening the budget alone to 25% of the FWHM, the 1 dB point of a Lorentzian drop filter, gives 25 and 32 ms, and 30% gives 25 and 27 ms. The stall depends only weakly on the budget because the detuning overshoots it several-fold. It is sensitive to the swing. Halving the temperature excursion gives 10 and 8 ms. The idle-power fraction matters less, since changing it from 6% to 25% of peak, which includes the measured 19%, changes the delay by under 15% for the median devices and by about a third for the least-loaded one. The loop is the other parameter the result hinges on, and Section 4.7 gives its full dependence.

5 RELATED WORK

Wafer- and panel-scale optical interconnects. Several efforts co-package photonics with compute and memory dies at wafer or panel scale to overcome the pin-density and reach limits of electrical I/O. TSMC’s COUPE integrates photonic engines at the die level [13] and its system-on-wafer platform integrates full-reticle compute dies and memory on one wafer [34], Bao et al. analyze the cost and benefit of waferscale silicon photonics [4], Celestial AI reports an interposer-assembly process with optical bridges [2], panel-scale reconfigurable photonic fabrics have been proposed [14], 3D-integrated chip-to-chip photonic links demonstrated [9], and photonic device platforms for AI accelerators surveyed [39]. Bernadskiy et al. evaluate MoE training on a 3D-integrated optical platform [5]. These works establish the architectural and bandwidth case for wafer- and panel-scale optics, largely under steady-state or synthetic traffic assumptions. This paper instead evaluates such a fabric under the workload-coupled thermal transients of real MoE training and shows that they materially affect the bandwidth these architectures deliver.

Photonic tuning and thermal management. The vulnerability of microring links to thermal drift is well documented at the device level, through wavelength-locking loop design [1, 10, 29, 38], thermal characterization of co-packaged optics [7, 8], and thermally aware transmitter design [33], which quantify loop bandwidth and cooling behavior for individual rings or small arrays under controlled or steady-state loads. Non-volatile alternatives to thermo-optic tuning [30, 37, 45] and athermal lithium-niobate resonators [11, 21] have been characterized separately. None of these studies connects device-level thermal and tuning behavior to the communication timing of large-scale distributed training, which Sections 4.2 to 4.6 do by deriving per-round stalls from a workload-driven thermal transient and feeding them back into the network timeline.

Communication scheduling and reconfigurable fabrics for MoE. At the network layer, MixNet reconfigures an optical-electrical fabric at runtime in response to MoE’s skewed all-to-all traffic [20], Chronos preschedules circuit-switched paths for LLM training [31], Morphlux reconfigures a torus fabric for multi-tenant ML jobs [17], InfiniteHBD builds a datacenter-scale high-bandwidth domain with optical circuit switching [36], and Janus replaces the token all-to-all with scheduled expert fetching [22]. The optical systems among these treat the reconfiguration or switching time of the optical layer as a fixed or assumed parameter rather than deriving it from the thermal behavior of the photonic devices, which is the gap our workload-to-thermal-to-optical-to-network methodology addresses.

This work is complementary to both lines and contributes the quantitative link between them. It quantifies, through a physically grounded and self-consistent chain from GPU power to ring detuning to communication stall, how much of a wafer-scale optical fabric’s bandwidth a conventional thermo-optic tuning loop delivers under MoE training, and what a faster loop, a predictive heater drive, a dummy load, or a drift-free ring would each have to deliver to recover the rest.

6 CONCLUSION

This work analyzes wafer-scale photonic interconnects for MoE training with a cross-layer methodology that chains packet-level network simulation, a transient thermal model grounded in published stack data and cross-checked against H100 die-temperature telemetry, and an optical-domain stall criterion, and feeds the resulting per-round stalls back into the network simulation. MoE execution produces 10 to 11 K temperature swings at the photonic layer between compute bursts and idle intervals. With a thermo-optic loop limited to the measured 5 nm/s, the stalls lengthen the training iteration of Mixtral 8×7B and LLaMA-MoE 6.7B by 1.58× and 2.82× in an open-loop estimate and by 1.32× and 1.55× once the stall is fed back, on a fabric whose stall-free iteration time is within 3% and 16% of an ideal non-blocking fat-tree. The thermal feedback reveals that the stall reshapes the duty cycle of each GPU and thereby its own thermal cause, and the stall persists at full model depth at about 1.16× and 1.25×. The analysis further turns the penalty into specifications. A loop that slews at 40 nm/s or reacts within 0.5 ms, a heater pre-driven within about 1 ms of each GPU kernel launch, or a dummy load holding the GPUs above 75% of peak power each removes it, and added thermal mass does not. Removing the drift at the ring is the least costly of these options. An athermalized lithium-niobate ring with a non-volatile ferroelectric setpoint keeps its residual inside the detuning budget if its athermal point lies within about 3 K of the operating temperature, and unlike the other options it needs no holding power, no fast loop, no signal from the GPU, and no tuning to the workload, whereas a ferroelectric actuator used to track the drift falls short on range and write endurance. For a wafer fabric provisioned well enough to approach a non-blocking network, the tracking capability of the tuning loop is a first-order requirement, together with a CPO port on every reticle and expert-parallel placement within a wafer.

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